Lecture 1

1. Probability:

definition of Probability: $P(A)$: probability of event A.

$$P(A) = \frac{N_A}{N}$$: N_A: number of occurences of A

N: number of all possible out comes

2. conditional Probability:

$P(A|B)$: The probability of event A given that event B is true.

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A,B)}{P(B)} \qquad P(B) \neq 0$$

$$\Rightarrow P(A,B) = P(A|B)P(B) = P(B|A)P(A)$$

3. Total Probability Theorem:

If $\{A_1, \cdots, A_n\}$ is a partision of A, B is arbitrary event:

$$P(A) = P(A_1, A_2, A_3, \cdots, A_n) = P(A_1) + \cdots + P(A_n), \quad P(A_i, A_j) = 0$$

$$P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \cdots + P(B|A_n)P(A_n)$$

$$= \sum_i P(B|A_i)P(A_i)$$

$$P(B) = P(B, A_1) + P(B, A_2) + \cdots + P(B, A_n)$$

$$= P(B|A_i)P(A_i) + \cdots\cdots + P(B|A_n)P(A_n)$$

$$= \sum P(B|A_i)P(A_i)$$

4. Bayes Theorem:

$$\frac{P(A_i B)}{} = \frac{P(B|A_i)P(A_i)}{}$$

$$P(A_i|B) = \frac{\quad}{P(B)} = \frac{\quad}{\sum_i P(B|A_i)P(A_i)}$$

Independence:

5. $P(A \text{ and } B) = P(A) + P(B) - P(A \text{ or } B)$

$$P(A,B) = P(A|B)P(B) = P(A)P(B)$$

$$= P(B|A)P(A) = P(B)P(A)$$

Random Variables:

A random variable defines a function that quantifies an experiment.

For example: $S = \{s_1, \cdots, s_n\}$, $x: S \to \mathbb{R}$, $x(s_i) = x_i$

$s_i \Rightarrow t_i$, $x_i : V_i$ $\quad V(t_i) = V_i$

$$P(t_i) = P_i$$

Definition: Distribution: The distribution of a random variable:

$$F_x(x) = P(x \le x)$$

$P(x \le x)$ represents the probability that x has values lower than x

Probability density function (pdf):

$$p(x) = \lim_{\varepsilon \to 0} \frac{F_x(x) - F_x(x-\varepsilon)}{\varepsilon} = \frac{dF_x(x)}{dx} = \frac{dP(x \le x)}{dx}$$

Hence: $P(x \le x) = \int_{-\infty}^{x} p(\varepsilon)\, d\varepsilon$

$$P(x \le \infty) = 1 = \int_{-\infty}^{+\infty} p(\varepsilon)\, d\varepsilon$$

$P(x_1 \le x < x_2) = \int_{x_1} p(\xi) d\xi$

example distributions:

Gaussian distributions: $x \sim N(m, \sigma) \Leftrightarrow p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-m)^2}{2\sigma^2}}$

Uniform distributions:

$$p(x) = \begin{cases} 0 & x < x_1 \\ \frac{1}{x_2 - x_1} & x_1 \le x \le x_2 \\ 0 & x > x_2 \end{cases}$$

conditional distribution: $P(\cancel{x} \le x \mid A) = \frac{P(X \le x, A)}{P(A)}$

conditionall probability dencity function:

$$p(x|A) = \frac{dP(X \le x \mid A)}{dx}$$

The Total probability theorem for pdf:

For: $A = \{A_1, \cdots, A_n\}$, $A_1, \cdots A_n$ are exclusive.

$$P(X \le x \mid A) = P(X \le x \mid A_1) P(A_1) + \cdots + P(X \le x \mid A_n) P(A_n)$$

$$p(x) = p(x|A_1) P(A_1) + \cdots + p(x|A_n) P(A_n)$$

Example:

$$p(x) = \frac{dP(X \le x)}{dx} \approx \frac{P(X \le x + \Delta x) - P(X \le x)}{\Delta x}$$

$$P(X = x) = P(X \le x + \Delta x) - P(X \le x) = p(x) \cdot \Delta x$$

$$P(X = x \mid A) = p(x|A) \cdot \Delta x$$

$$P(A \mid X = x) = \frac{P(A, X = x)}{P(X = x)} = \frac{P(X = x \mid A) P(A)}{P(X = x)} = \frac{p(x|A)}{p(x)} \cdot P(A)$$

$$\Rightarrow P(A \mid x=x)\, p(x) = p(x \mid A)\, P(A)$$

continuous theorem:

$$P(A) = \int_{-\infty}^{+\infty} P(A \mid x=x)\, p(x)\, dx$$

Continuous Bayes' theorem:

$$p(x \mid A) = \frac{P(A \mid x=x)}{P(A)} \cdot p(x) = \frac{P(A \mid x=x)\, p(x)}{\int_{-\infty}^{+\infty} P(A \mid x=x)\, p(x)\, dx}$$

Mean: The expected value or mean of a continuous variable is:

$$m_x = E(x) = \int_{-\infty}^{+\infty} x\, p(x)\, dx$$

Conditional Mean:

$$m_x = E(x \mid A) = \int_{-\infty}^{+\infty} x\, p(x \mid A)\, dx$$

Variance:

$$\sigma_x^2 = E\left((x-m_x)^2\right) = \int_{-\infty}^{+\infty} (x-m_x)^2\, p(x)\, dx$$

$$= E(x^2) - (E(x))^2$$

Conditional variance:

$$\sigma_{x|A}^2 = E\{(x-m_{x|A})^2 \mid A\} = \int_{-\infty}^{+\infty} (x-m_{x|A})^2\, p(x \mid A)\, dx$$

A set of random variables:

Joint probability density function:

$x_1, \cdots, x_n$, $\quad p(x_1, \cdots, x_n) = \frac{d\, F(x_1, \cdots, x_n)}{}$

$dx_1, \cdots dx_n$

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} p(x_1, \cdots, x_n)\, dx_1 \cdots dx_n = 1$$

if $x_i \cdot x_j$, are independent, $p(x_1, x_2) = p(x_1) p(x_2)$

For: $y = \alpha x_i + \beta x_j \Rightarrow E(y) = \alpha E(x_i) + \beta E(x_j)$

covariance for σ_{ij}: $\sigma_{ij}^2 = E((x_i - m_i)(x_j - m_j))$

$$= E(x_i x_j) - E(x_i) E(x_j)$$

Hence: $\sigma_{ij} = 0$, $E(x_i x_j) = E(x_i) E(x_j)$

they are uncollerated.

① If x_i, x_j are independent, $p(x_i, x_j) = p(x_i) p(x_j) \Rightarrow E(x_i x_j) = E(x_i) E(x_j)$

② If x_i, x_j are uncorrelated, are they independent?

③ if x_i, x_j subject to Gaussian distribution, independent $\overset{?}{=}$ uncorrelated?

Then random vector

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

All the element are random variables

$E(X) \hat{=} m_X$

$$P = E((X - m_X)(X - m_X)^T) = E(XX^T) - E(X) E(X^T)$$

$$P = \begin{bmatrix} P_{11} & \cdots & P_{1n} \\ \vdots & & \vdots \\ P_{n1} & \cdots & P_{nn} \end{bmatrix}$$

$P_{ii} = E(x_i - m_{x_i})^2 = \int_{-\infty}^{+\infty} (x_i - m_{x_i}) p(x_i)\, dx_i$

$P_{ij} = E((x_i - m_{x_i})(x_j - m_{x_j}))$

$= P_{ji}$

$$\begin{cases} P \text{ is symmetric Matrix} \\ P \text{ is positive definite Matrix.} \end{cases}$$

Gaussian Vector pdf: $p(X) = \frac{1}{(2\pi)^{\frac{n}{2}} \sqrt{|P|}} e^{-\frac{1}{2}(X-m_x)^T P^{-1} (X-m_x)}$

$$\begin{cases} |P| \text{ is the determinat of } P \\ m_x = E(X) \\ P = E((X-m_x)(X-m_x)^T) \end{cases}$$

A homework: $X = [x_1, \cdots, x_n]^T$

$p(X) = N(m_x, P_{xx})$

define $q = (X-m_x)^T P_{xx}^{-1} (X-m_x)$ $\quad y = P_{xx}^{-\frac{1}{2}} (X-m_x)$

$= y^T y = \Sigma y_i^2$

what's mean and covariance for q?

Maximum Likely hood and max. Posterior Estimator:

$$\begin{cases} \max p(z|x) : \text{MLE} \\ \max p(x|z) \quad \text{MPE} \end{cases}$$

$p(z|x)$

$p(x|z) = \frac{p(z|x)\,p(x)}{p(z)}$

Gaussian distribution:

$z_i = h_i(x) + n_i \Rightarrow \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} h_1(x) \\ h_2(x) \\ \vdots \\ h_1(x) \end{bmatrix} + \begin{bmatrix} n_1 \\ \vdots \\ n_n \end{bmatrix}$

$$z = h(x) + n$$

$$\max_x p(z|x) = \max_x (p(z_1, \cdots, z_n|x))$$

$$= \max\left(\frac{1}{(2\pi)^{\frac{n}{2}} |P|} \exp(-(z-h(x))^T P^{-1} (z - h(x)) \right)$$

$$\max_x (p(z|x)) \Rightarrow \min_x (z - h(x))^T P^{-1} (z - h(x)) = C$$

$$z = h(x) + n = h(\hat{x}) + H(x - \hat{x}) + n$$

$$z - h(\hat{x}) = H(x - \hat{x}) + n \qquad \tilde{z} = z - h(\hat{x}) \quad x - \hat{x} = \tilde{x}$$

$$\min C \approx \min(\tilde{z} - H\tilde{x})^T P^{-1} (\tilde{z} - H\tilde{x})$$

$$= \min(\tilde{z}^T \tilde{z} - 2 \tilde{z}^T P^{-1} H \tilde{x} + \tilde{x} H^T P^{-1} H \tilde{x})$$

$$H^T P^{-1} H \boxed{\tilde{x}} = H^T P^{-1} \tilde{z} \quad \Rightarrow \quad x^{\oplus} = \hat{x} + \tilde{x} = \hat{x} + (H^T P^{-1} H)^{-1} H^T P^{-1} \tilde{z}$$

This is the classical weighted least squares

What's the covariane ~~of~~ the new estimates: $x^{\oplus}$?

What's the MP?

$$\begin{cases} x_p = x + n_p \\ z_i = h_i(x) + n_i \quad i = 1, \cdots, n \end{cases}$$

Then, what's the solution for x of $\max p(x|z)$

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optimization: $\begin{cases} \text{Gaussian Newton (GN)} \\ \text{leverberg marquart (LM)} \end{cases}$

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linear system:

Linear system

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \qquad \begin{cases} x: \text{ is the state} \\ u: \text{ input} \\ y: \text{ output} \end{cases}$$

This is a continuous-time system:

$$\begin{cases} x_{k+1} = A' x_k + B' u_k \\ y_{k+1} = C' x_{k+1} + D' u_{k+1} \end{cases}$$ This a discrete-time system.

If, A, B, C, D are constant, linear time invariant system (LTI)

If A, B, C, D are depent ont time, linear time variant system (LTV)

Let's start from a simple system:

$$\begin{cases} \dot{x} = Ax, \text{ with } x(t_0) = x_0 \\ y = Cx \end{cases}$$

Solution for $\dot{x} = Ax$ with $(x(t_0)) = x_0$

$\dot{x} - Ax = 0$ $\qquad \dfrac{d(e^{-At}x)}{dt} = e^{-At}\dot{x} + e^{-At}(-A)x = e^{-At}(\dot{x} - Ax) = 0$

$$\int_0^t \frac{d(e^{-A\tau}x)}{d\tau} = 0 = e^{-A\tau}x\Big|_0^t = e^{-At}x(t) - e^{-A0}\cdot x(t_0) = e^{-At}x(t) - x(t_0)$$

$x(t) = e^{At}x_0$ $\quad$. $x(t) = \Phi(t,0)\, x_0$ $\quad \Phi(t,0) = e^{At}$ state transition matrix.

$y = C \cdot e^{At} x_0$

Hence: the fina solution: $\begin{cases} x(t) = \Phi(t,0)\, x_0 \\ y = C\,\Phi(t,0)\, x_0 \end{cases}$, $\Phi(t,0) = e^{At}$

What's the property of state transition matrix?

$\Phi(t_2, t_0) = \Phi(t_2, t_1)\, \Phi(t_1, t_0)$

$\frac{d}{dt}\Phi(t,0) = A\,\Phi(t,0)$

$\dot{x} = Ax \qquad x = \Phi(t,0)x_0 \qquad \dot{x} = \dot{\Phi}(t,0)x_0 = Ax = A\Phi(t,0)x_0$

Find the correct prove for this.

Then: with $x(t) = \Phi(t,0)x_0$

$y = Cx$

t_0: $x_0 \quad y_0 = Cx_0 \qquad y_0 = Cx_0$

t_1: $x_1 = \Phi(1,0)x_0 \quad y_1 = Cx_1 \qquad y_1 = Cx_1 = C\Phi(1,0)x_0$

t_2: $x_2 = \Phi(2,1)x_1 \quad y_2 = Cx_2 \qquad y_2 = Cx_2 = C\Phi(2,0)x_0$

$\vdots \qquad \vdots \qquad \vdots$

$t_n = \quad x_n = \Phi(n,n-1)x_{n-1} \quad y_n = Cx_n$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ C\Phi(1,0) \\ \vdots \\ C\Phi(2,0) \end{bmatrix}}_{\mathcal{O}} x_0$$

Now let's consider the whole system:

$\dot{x} = A(t)x + B(t)u(t)$ solve this linear system.

$y = C(t)x + D(t)u(t)$

$\dot{x} = A(t)x + B(t)u(t) \Rightarrow \quad \dot{x} - A(t)x = B(t)u(t)$

$$\frac{d\left[e^{-\int_0^t A(\tau)d\tau}x\right]}{dt} = e^{-\int_0^t A(\tau)d\tau}\dot{x} - e^{-\int_0^t A(\tau)d\tau}A(t)\cdot x$$

$$= e^{-\int_0^t A(\tau)d\tau}\underbrace{(\dot{x} - A(t)x)}$$

$$= e^{-\int_0^t A(\tau)d\tau}B(t)u(t)$$

$$e^{-\int_0^t A(\tau)d\tau}x\Big|_0^t = e^{-\int_0^t A(\tau)d\tau}x(t) - e^{-\int_0^0 A(\tau)d\tau}x_0$$

$$= e^{-\int_0^t A(\tau)d\tau} x(t) - x_0$$

$$= \int_0^t e^{-\int_0^\tau A(s)ds} B(\tau)u(\tau)d\tau$$

$$x(t) = \underbrace{e^{\int_0^t A(\tau)d\tau}}_{\Phi(t,0)} x_0 + \int_0^t \underbrace{e^{\int_0^{t-\tau} A(s)ds}}_{\Phi(t,\tau)} B(\tau)u(\tau)d\tau$$

$$x(t) = \Phi(t,0)x_0 + \int_0^t \Phi(t,\tau)B(\tau)u(\tau)d\tau$$

$$y = C(t)x(t) + D(t)u(t)$$

$$= C(t)\left(\Phi(t,0)x_0 + \int_0^t \Phi(t,\tau)B(\tau)u(\tau)d\tau\right) + D(t)u(t)$$

This is one of the most important linear system properties.